

Project Proposal

StudyBuddy

AI-Powered Educational Planning Platform — A Personal AI Assistant for Exam Preparation, Study Roadmaps & Smart Reminders

Members	Yash Ahir , Daksh Digashia
Roll Number / Enrollment ID	D25CS118 , D25CS124 (Batch C-2)
Institute	CSPIT, Charusat University

Problem Definition Statement

Students preparing for school, college, or competitive exams struggle to plan an effective, personalised study schedule. Exam dates, syllabi, and study material — handwritten notes, PDFs, YouTube lectures/playlists — are scattered across different places, and manually working out what to study, when, and for how long is time-consuming and error-prone. This often leads to last-minute cramming, missed topics, and rising exam stress. There is no single assistant that turns a student's exam details and existing material into a structured, adaptive study roadmap while also letting them directly ask questions about their own notes and videos.

Product Overview

1.1 Users

Student (Primary User)

- Upload exam/test dates along with the syllabus (typed, PDF, or scanned photo)
- Upload notes, PDFs, and YouTube videos/playlists as study material
- Receive an AI-generated, day-wise study roadmap synced to a calendar
- Get daily reminders and adaptive rescheduling when a session is missed
- Ask questions about a specific part of a video or notes and get sourced answers
- Take auto-generated quizzes and review auto-generated flashcards

Parent / Guardian

- Optional read-only access to a student's roadmap and progress dashboard
- Receive shared progress summaries ahead of exams

Additional stakeholder: the platform can later expose an educator/admin view for institutions that want to track cohort-level exam readiness. Clear separation between the student's working view and the parent's monitoring view keeps the core experience simple.

1.2 Functionality

- Accept exam/test dates and syllabus, typed or uploaded as a photo/PDF (OCR-parsed)
- Accept study material in multiple formats: notes, PDFs, and YouTube videos/playlists
- Use the Google Gemini API to parse the syllabus and generate a day-wise study roadmap
- Automatically schedule study sessions on an in-app calendar and send daily reminders
- Answer questions about specific parts of a video or notes using a Retrieval-Augmented Generation (RAG) pipeline
- Automatically generate notes, quizzes, and flashcards from uploaded content
- Adaptively reschedule the roadmap when a session is missed or a topic is marked difficult

1.3 Features

- Adaptive Re-scheduling — shifts the roadmap automatically based on missed sessions or topic difficulty
- Topic Weightage Detection — prioritises high-yield topics using syllabus and past-paper analysis
- YouTube Timestamp Q&A — jumps to the exact moment in a video using transcript-grounded answers

- Auto-Generated Flashcards & Mind Maps — spaced-repetition cards and visual summaries from notes
- Auto-Generated Quizzes — topic-wise quizzes with performance tracking
- Handwritten Doubt Solver — photo-based doubt solving via OCR
- Progress Analytics Dashboard — syllabus completion %, weak vs. strong topics
- Gamification — study streaks, XP, badges, and friend leaderboards
- Multilingual Support — doubt-solving in Hindi/Gujarati and text-to-speech for notes

Proposed Solution

A cross-platform personal AI assistant where a student uploads exam dates, syllabus, and study material, and the AI plans, schedules, and reinforces learning end-to-end — replacing scattered notice-board-style planning with one adaptive, centralised roadmap.

High-Level Architecture

User (Flutter app) → Node.js / Express API → Google Gemini API + MongoDB Atlas

- AI layer: Google Gemini API (roadmap generation, quizzes, doubt-solving)
- Retrieval: MongoDB Atlas Vector Search (RAG for notes/video Q&A)
- Auth & notifications: Firebase Authentication + Cloud Messaging
- File storage: Cloudinary
- Scheduling: node-cron for daily reminder jobs

Technology Stack

Frontend	Backend	AI, Data & Deployment
Flutter (Android, iOS, Web)	Node.js, Express, JWT	Google Gemini API, MongoDB Atlas
Provider / Riverpod	node-cron, Firebase Admin	Atlas Vector Search, Firebase Auth + FCM
table_calendar, fl_chart	Multer, PyMuPDF/pdf-parse	Cloudinary, Render (backend), Vercel (web)

Highlight Feature — RAG-Based Video & Notes Q&A

Uploaded notes and video transcripts are chunked and converted into vector embeddings using the Gemini Embeddings API, then stored in MongoDB Atlas Vector Search. When a student asks a question, the system retrieves only the most relevant chunks and passes them to Gemini, so answers are grounded in the student's own material — including jumping to the exact timestamp in a YouTube video — rather than generic, ungrounded responses.

Expected Outcome & Future Plan

3.1 Expected Outcome

- A working cross-platform prototype where a student uploads a syllabus and exam date and receives an automatically generated, day-wise study roadmap
- Automated calendar scheduling with daily reminders, reducing manual planning effort
- A functional content Q&A system that answers questions from uploaded notes and YouTube video timestamps
- Auto-generated quizzes and flashcards that reinforce learning and measure topic-wise readiness
- A measurable reduction in the time students spend manually organising exam preparation

Future Plan / Roadmap

- Collaborative Study Rooms — shared roadmaps and leaderboards for peers preparing for the same exam
- Previous-Year Question Pattern Analysis — lightweight scikit-learn classifier trained in Google Colab to predict high-yield topics
- Offline / Low-Connectivity Mode — local caching of notes and roadmap for uninterrupted access
- Institution Pilot — deploy with a real batch of students to gather feedback and iterate

Next steps: prototype demo → usability testing with classmates → pilot rollout for one semester's exam cycle.

Challenges

- Free-tier API rate limits (Gemini, YouTube Data API) may restrict usage during heavy concurrent testing
- OCR accuracy for handwritten syllabus/doubt images is limited compared to typed text
- Fetching accurate transcripts for YouTube videos without captions requires extra processing (e.g., Whisper-based transcription)
- Designing an adaptive scheduling algorithm that meaningfully reprioritises topics without becoming rigid or repetitive
- Managing free-tier hosting limitations, such as backend cold-starts on Render, without affecting user experience
- Keeping the RAG pipeline's retrieved context accurate and relevant as the volume of uploaded notes grows